

Jeffrey Wang

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EDUCATION

Carnegie Mellon University (CMU) | Bachelor of Computer Science and Art (BCSA) | GPA: 3.75/4.00

May 2027

EXPERIENCE

Software Developer Intern | Carnegie Mellon University

May 2026 - Present

- Developed a mobile campus exploration platform for CMU Orientation, replacing staff-mediated check-ins with self-service QR/NFC and geolocation verification across web, iOS, and Android using React, TypeScript, Tauri, Rust, and PostgreSQL.
- Scaled platform to 1,000+ concurrent users reducing p95 API latency by 80% to 113 ms by eliminating N+1 queries and utilizing two-tier caching with a 98.77% cache-hit rate while protecting concurrent traffic with atomic PostgreSQL transactions.

Software Developer Intern | CMU Center for Transformational Play

May 2025 - December 2025

- Shipped a Unity mobile application for Allegheny Health Network with a 12-person cross-functional team, delivering weekly production builds for 2 deployment studies.
- Developed a Python CI/CD pipeline with Myers' diff-based selective Unity content regeneration, cutting content integration time from 10 minutes to under 1 minute.
- Profiled Unity render passes with RenderDoc frame captures, reducing GPU runtime 50% on lower-end devices.

Software Developer Intern | CMU Software Engineering Institute

October 2024 - May 2025

- Architected a modular Unity/C# assessment engine for Air Force data-science hiring, decoupling simulation, scoring, and task-selection into independent systems that supported 50+ assessment scenarios without core rewrites.
- Developed an adaptive assessment pipeline selecting from 20 simulation scenarios by skill, difficulty, and real-time performance; prototyped Bayesian ability scoring across accuracy and reaction time via 5 subject-matter experts in a vertical-slice pilot.
- Engineered an offline-first telemetry and persistence system with JSON-based storage and event-driven Unity components, preserving assessment results across 3-hour sessions without network access and auto-syncing on reconnect.

President | Game Creation Society

May 2024 - Present

- Led weekly game-development workshops for 100+ members and mentored project leads managing teams of 4-30+ contributors.
- Organized annual game showcases attracting hundreds of visitors and featuring student-led games and interactive projects.

PROJECTS

Vulkan Render

January 2026 - Present

- Built a real-time Vulkan renderer in C++ with scene graph loading, scene viewer, and dynamic point, spot, and directional lighting.
- Implemented a physically based shading pipeline with GGX materials and split-sum image-based lighting using compute-generated irradiance maps, prefiltered environment maps, and BRDF LUTs; added shadow mapping for spot lights and PCSS soft shadows.
- Prototyped a real-time radiance-cache global illumination, combining screen-space probes with a world-space probe cache to approximate indirect lighting in dynamic scenes.

Monte-Carlo Ray Tracing

January 2026 - May 2026

- Implemented Specular Manifold Sampling in C++ to solve difficult mirror and glass caustics with Newton-based nonlinear optimization; improved solver reliability 4.76x through guided initialization.
- Parallelized and optimized the renderer with OpenMP, cutting single-thread solver time 53.6%, reaching 3.84x speedup on 8 CPU threads, and adding 13 deterministic rendering/multithreading regression tests.

CMU Maps (Room-level Navigation Application)

May 2024 - Present

- Designed a room-level navigation PWA(React, Node.js, PostgreSQL) serving 1000+ daily concurrent users, converting 200+ floor plans into connected routing graphs for multi-floor, elevator-accessible, and indoor-preferred navigation.
- Coordinated with CMU administration and Spring Carnival organizers to gather requirements and location data, integrating 125+ Spring Carnival events into CMU Maps for search and navigation used during an event serving 5,000+ attendees

C++ Soft-body Game Engine

August 2025 - December 2025

- Designed a C++ soft-body physics engine with real-time OpenGL mesh updates and metaball implicit-surface rendering in GLSL for smooth pooling and melting VFX.
- Implemented an OpenGL/GLSL pipeline integrating forward lighting, shadow-atlas PCF, metaball ray marching, and HDR bloom

SKILLS

Languages: C, C++, Python, TypeScript, JavaScript, C#, SQL, Rust, HTML/CSS

Graphics & Game: Vulkan, OpenGL, GLSL, HLSL, Unity, Unreal Engine, Real-Time Rendering

Tools: React, React Native, Node.js, PostgreSQL, SQL, REST APIs, Docker, AWS, Oracle, Git, GitHub, CI/CD, Agile